

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET E — CASE STUDY & PRACTICAL EXPERIMENTS

CHAPTER 01: EXPLORATION: ENTERING THE WORLD OF SECONDARY SCIENCE
TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-E-01

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET E - CASE STUDY & PRACTICAL EXPERIMENTS):

1. The question paper comprises **5 In-Depth Practical Case Studies** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each laboratory experiment, data table, and passage carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: SIMPLE PENDULUM OSCILLATION & DETERMINATION OF ACCELERATION DUE TO GRAVITY (\$g\$)

A group of Class 9 students set up a simple pendulum in the physics laboratory to study the variation of its time period (\$T\$) with its effective length (\$L\$). They used a light, inextensible thread tied to a spherical brass bob of mass 50 g. The effective length was measured from the point of suspension to the center of gravity of the bob. Keeping the angular displacement small (\$< 10^\circ\$), they timed 20 complete oscillations using a digital stopwatch for different lengths and recorded the data below:

Table 1: Pendulum Experimental Observations

- Length \$L_1 = 0.40\text{ m} \rightarrow\$ Time for 20 oscillations = \$25.4\text{ s} \rightarrow T = 1.27\text{ s} \rightarrow T^2 = 1.61\text{ s}^2\$
- Length \$L_2 = 0.60\text{ m} \rightarrow\$ Time for 20 oscillations = \$31.1\text{ s} \rightarrow T = 1.55\text{ s} \rightarrow T^2 = 2.40\text{ s}^2\$
- Length \$L_3 = 0.80\text{ m} \rightarrow\$ Time for 20 oscillations = \$35.9\text{ s} \rightarrow T = 1.80\text{ s} \rightarrow T^2 = 3.24\text{ s}^2\$
- Length \$L_4 = 1.00\text{ m} \rightarrow\$ Time for 20 oscillations = \$40.1\text{ s} \rightarrow T = 2.01\text{ s} \rightarrow T^2 = 4.04\text{ s}^2\$

The students then plotted a graph of \$T^2\$ versus \$L\$ to calculate \$g\$ using the relation \$T = 2\pi \sqrt{L/g}\$.

Q.1 What is the expected nature and mathematical shape of the graph plotted between \$T^2\$ (on y-axis) and length \$L\$ (on x-axis)?

- | | |
|---|--|
| <p>(A) A parabola curving towards the length axis</p> <p>(C) A hyperbola curving asymptotically</p> | <p>(B) A straight line passing through the origin</p> <p>(D) A horizontal line parallel to the length axis</p> |
|---|--|

Q.2 Based on the experimental data for $L = 1.00\text{ m}$ ($T^2 = 4.04\text{ s}^2$), what is the calculated experimental value of g (take $\pi = 3.1416$)?

- (A) 9.77 m/s^2 (B) 9.80 m/s^2
(C) 10.2 m/s^2 (D) 9.50 m/s^2

Q.3 Why were the students instructed to record the time for 20 full oscillations rather than timing a single oscillation?

- (A) Because a pendulum bob stops after 20 oscillations (B) To substantially reduce human reaction time error in stopwatch operation by distributing it over 20 cycles
(C) Because the formula is valid only for 20 oscillations (D) To increase the amplitude of the pendulum

Q.4 If the 50 g brass bob is replaced by a 150 g lead bob while keeping the length strictly at 1.00 m, what will happen to the time period?

- (A) It will triple to 6.03 s (B) It will decrease to 1.16 s
(C) It will remain unchanged at 2.01 s (D) It will increase by $\sqrt{3}$ times

Q.5 Why is it essential to maintain small angular amplitude ($< 10^\circ$) during simple pendulum experiments in school laboratories?

- (A) To prevent air resistance from breaking the string (B) Because simple harmonic motion is strictly valid under the small angle approximation $\sin\theta \approx \theta$
(C) To allow the bob to accelerate past the speed of sound (D) To increase the value of gravitational acceleration

CASE STUDY 2: PRECISION METROLOGY WITH VERNIER CALIPERS AND SCREW GAUGE

In a physics metrology practical, students measured the dimensions of a solid metallic cylinder and a thin copper wire using precision laboratory instruments.

- Instrument 1 (Vernier Calipers):** Used to measure the length and external diameter of the cylinder. The main scale is graduated in millimeters (1 MSD = 1 mm). 10 vernier scale divisions (VSD) coincide exactly with 9 main scale divisions.
- Instrument 2 (Micrometer Screw Gauge):** Used to measure the diameter of the thin copper wire. The pitch of the screw is 0.5 mm, and the circular thimble has 50 equal divisions.

Before taking readings, the students inspected both instruments for zero error. The screw gauge showed that when the studs were brought into contact, the zero mark of the circular scale was 3 divisions below the reference line.

Q.6 What is the least count of the Vernier Calipers used by the students?

- (A) 0.1 cm (B) 0.01 cm (0.1 mm)
(C) 0.001 cm (D) 0.05 mm

Q.7 What is the least count of the Micrometer Screw Gauge?

- (A) 0.01 mm (B) 0.001 mm
(C) 0.05 mm (D) 0.1 mm

Q.8 What type of zero error is present in the screw gauge when its circular zero lies 3 divisions below the reference line?

- (A) Positive zero error of $+0.03\text{ mm}$ (B) Negative zero error of -0.03 mm
(C) Positive zero error of $+0.15\text{ mm}$ (D) Zero error is absent

Q.9 While measuring the diameter of the wire, the main scale reads 1.5 mm and the 38th circular division aligns with the reference line. What is the true corrected diameter of the wire?

- (A) 1.88 mm (B) 1.85 mm
(C) 1.91 mm (D) 1.53 mm

Q.10 Why is a ratchet provided at the back of a micrometer screw gauge?

- (A) To quickly spin the screw without measuring (B) To prevent overtightening and ensure uniform contact pressure on the object during all measurements
(C) To lock the main scale firmly in place (D) To calibrate the pitch of the screw

CASE STUDY 3: DENSITY DETERMINATION & ARCHIMEDES' LIQUID DISPLACEMENT

A student was provided with an irregular piece of solid stone of mass 54.0 g. To determine its density experimentally, she used a spring balance, a graduated measuring cylinder of capacity 100 mL, and water of density 1.00 g/cm^3 .

- The spring balance had a least count of 1.0 g.
- The initial water volume in the graduated cylinder was recorded as $V_1 = 50.0 \text{ mL}$.
- The stone was tied to a fine thread and gently lowered completely into the water without touching the walls or bottom of the cylinder.
- The new water level rose to $V_2 = 70.0 \text{ mL}$.

The student then repeated the immersion using a saturated salt brine solution of density 1.20 g/cm^3 .

Q.11 What is the volume of the irregular stone piece?

- (A) 50.0 cm^3 (B) 70.0 cm^3
(C) 20.0 cm^3 (D) 120.0 cm^3

Q.12 What is the calculated density of the stone in SI units (kg/m^3)?

- (A) 2.70 kg/m^3 (B) 2700 kg/m^3
(C) 270 kg/m^3 (D) 0.27 kg/m^3

Q.13 What is the magnitude of the buoyant force exerted by pure water on the completely submerged stone ($g = 9.8 \text{ m/s}^2$)?

- (A) 0.540 N (B) 0.196 N
(C) 0.200 N (D) 1.96 N

Q.14 When the same stone is immersed in the salt brine (density 1.20 g/cm^3), what is the new buoyant force exerted on it?

- (A) 0.196 N (B) 0.235 N
(C) 0.163 N (D) 0.270 N

Q.15 Which precautionary technique ensures accurate volume reading of the water meniscus in a glass graduated cylinder?

- (A) Reading the highest edge of the convex liquid curve (B) Viewing the cylinder from an oblique angle above the scale
(C) Placing the eye level exactly horizontal to the lowest point of the concave meniscus (D) Adding oil to flatten the meniscus curve

CASE STUDY 4: THERMAL EXPANSION OF METALS & BIMETALLIC MECHANICS

In an experiment investigating the thermal expansion of solids, a student heated three metallic strips of equal initial length (100.0 cm) and cross-section from 25°C to 125°C (temperature increase $\Delta T = 100^\circ\text{C}$). The expansions recorded were:

- Strip A (Copper): $\Delta L = 1.70\text{ mm}$
- Strip B (Iron): $\Delta L = 1.20\text{ mm}$
- Strip C (Aluminum): $\Delta L = 2.40\text{ mm}$

The teacher then presented a bimetallic strip made by welding Strip A (Copper) and Strip B (Iron) together and heated it over a Bunsen burner flame.

Q.16 Which of the three metals exhibits the lowest coefficient of linear expansion (α)?

- | | |
|------------------------|---|
| (A) Strip A (Copper) | (B) Strip B (Iron) |
| (C) Strip C (Aluminum) | (D) All three have identical coefficients |

Q.17 What is the coefficient of linear expansion of Copper (Strip A) per degree Celsius?

- | | |
|---|---|
| (A) $1.7 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ | (B) $2.4 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ |
| (C) $1.2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ | (D) $1.7 \times 10^{-3} \text{ } ^\circ\text{C}^{-1}$ |

Q.18 When the Copper-Iron bimetallic strip is heated strongly, in which direction does it bend?

- | | |
|--|--|
| (A) Towards the Copper side | (B) Towards the Iron side |
| (C) It twists into a coil but remains straight | (D) It expands symmetrically without any curvature |

Q.19 If the same Copper-Iron bimetallic strip is subsequently cooled below room temperature in dry ice, what will occur?

- | | |
|--|--------------------------------------|
| (A) It bends towards the Iron side even more | (B) It bends towards the Copper side |
| (C) It fractures immediately into two pieces | (D) It stays completely straight |

Q.20 Which common home electrical appliance uses the bending of a bimetallic strip to automatically regulate temperature?

- | | |
|---|----------------------------------|
| (A) Electric kettle thermostat | (B) Incandescent filament bulb |
| (C) Electric immersion rod without controller | (D) Step-down voltage stabilizer |

CASE STUDY 5: CHEMICAL SAFETY & HAZARDOUS SPILL MANAGEMENT

During a chemistry laboratory orientation session, the lab instructor conducted a safety drill on handling hazardous chemicals, heating glassware, and responding to accidental chemical spills.

- **Scenario A:** A student accidentally spills concentrated sulfuric acid (H_2SO_4) onto the laboratory workbench and his forearm.
- **Scenario B:** Another student needs to prepare 250 mL of dilute sulfuric acid from concentrated acid.
- **Scenario C:** A liquid in an open glass beaker is being heated over a Bunsen burner.

Q.21 What is the correct and immediate first-aid response when concentrated acid spills onto a student's skin?

- | | |
|---|---|
| (A) Apply strong sodium hydroxide solution immediately to neutralize it | (B) Wash immediately under a continuous stream of copious running tap water for at least 15 minutes |
| (C) Smear the skin thickly with petroleum jelly or cooking oil | (D) Wrap the skin tightly with dry cotton bandage |

Q.22 What is the strictly mandated procedure for safely diluting concentrated sulfuric acid with water?

- (A) Pour water rapidly into concentrated acid while stirring (B) Add concentrated acid slowly and dropwise to water along the sides of the container with continuous stirring
- (C) Mix equal volumes of water and acid together in a sealed bottle and shake vigorously (D) Boil the acid first before mixing with cold water

Q.23 Why is a wire gauze placed on the tripod stand beneath a glass beaker during heating over a Bunsen burner?

- (A) To prevent chemical vapors from escaping into the room (B) To distribute flame heat evenly across the bottom of the beaker and prevent thermal shock breakage
- (C) To act as a chemical catalyst for the boiling reaction (D) To absorb smoke produced by the Bunsen flame

Q.24 Which region of a Bunsen burner flame is the hottest and recommended for efficient heating?

- (A) The inner dark cone of unburnt gas (B) The outer non-luminous pale blue zone of complete combustion
- (C) The luminous yellow zone of incomplete combustion (D) The base of the burner collar

Q.25 Which of the following personal protective equipment (PPE) is mandatory whenever heating chemicals or working with acids in the lab?

- (A) Safety goggles and cotton lab coat (B) Woolen gloves and dark sunglasses
- (C) Synthetic nylon jacket and flip-flops (D) Leather welding helmet

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & PRACTICAL INVESTIGATION EXPLANATIONS (SET E)

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.1	(B)	Squaring $T = 2\pi \sqrt{L/g}$ gives $T^2 = \left(\frac{4\pi^2}{g}\right) L$. Since $\frac{4\pi^2}{g}$ is a constant slope (m), the equation fits $y = mx$, which is a straight line passing through the origin.
Q.2	(A)	$g = \frac{4\pi^2 L}{T^2} = \frac{4 \times (3.1416)^2 \times 1.00}{4.04} = \frac{39.478}{4.04} \approx 9.77 \text{ m/s}^2$.
Q.3	(B)	Human reaction time error in pressing start/stop is typically ± 0.2 s. Timing 20 oscillations divides this error by 20, improving experimental accuracy.
Q.4	(C)	The time period of a simple pendulum is strictly independent of the mass and material of the bob ($T = 2\pi \sqrt{L/g}$ contains no mass term).
Q.5	(B)	The formula $T = 2\pi \sqrt{L/g}$ is derived assuming simple harmonic motion, which requires $\sin\theta \approx \theta$ in radians (valid only when $\theta < 10^\circ$).
Q.6	(B)	$\text{Least count} = 1 \text{ MSD} - 1 \text{ VSD} = 1 \text{ mm} - 0.9 \text{ mm} = 0.1 \text{ mm} = 0.01 \text{ cm}$.
Q.7	(A)	$\text{Least count} = \frac{\text{Pitch}}{\text{Number of circular divisions}} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$.
Q.8	(A)	When the zero mark of the circular scale is below the reference datum line, the zero error is positive ($+3 \times 0.01 \text{ mm} = +0.03 \text{ mm}$). It must be subtracted from the observed reading.
Q.9	(B)	Observed reading $= 1.5 \text{ mm} + (38 \times 0.01 \text{ mm}) = 1.88 \text{ mm}$. True reading $= \text{Observed} - (\text{Zero Error}) = 1.88 - (+0.03) = 1.85 \text{ mm}$.
Q.10	(B)	The ratchet slips and produces a clicking sound once optimum contact pressure is reached, preventing mechanical deformation of the object or stripping of the fine threads.
Q.11	(C)	Volume of stone $= V_2 - V_1 = 70.0 \text{ mL} - 50.0 \text{ mL} = 20.0 \text{ mL} = 20.0 \text{ cm}^3$.
Q.12	(B)	Density $\rho = \frac{\text{Mass}}{\text{Volume}} = \frac{54.0 \text{ g}}{20.0 \text{ cm}^3} = 2.70 \text{ g/cm}^3$. In SI units: $2.70 \times 1000 = 2700 \text{ kg/m}^3$.
Q.13	(B)	Buoyant force $F_b = V \rho_{\text{water}} g = (20 \times 10^{-6} \text{ m}^3) \times 1000 \text{ kg/m}^3 \times 9.8 \text{ m/s}^2 = 0.196 \text{ N}$.
Q.14	(B)	$F_b = V \rho_{\text{brine}} g = (20 \times 10^{-6}) \times 1200 \times 9.8 = 0.2352 \text{ N} \approx 0.235 \text{ N}$.
Q.15	(C)	Water forms a concave meniscus in glass due to adhesive forces. To avoid parallax error, the observer's line of sight must align horizontally with the bottom of the meniscus.
Q.16	(B)	Since $\Delta L = L_0 \alpha \Delta T$, for identical initial length and temperature change, Iron expanded the least (1.20 mm), meaning it has the smallest α .
Q.17	(A)	$\alpha = \frac{\Delta L}{L_0 \Delta T} = \frac{0.00170 \text{ m}}{1.00 \text{ m} \times 100^\circ \text{C}} = 1.7 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$.
Q.18	(B)	Copper expands more than Iron ($1.7 > 1.2$). To accommodate the greater length, Copper forms the outer curve, forcing the strip to bend towards the less-expanded Iron side.
Q.19	(B)	Upon cooling, Copper contracts more than Iron, forming the shorter inner curve. Hence, the strip reverses its bend and curves towards the Copper side.
Q.20	(A)	Thermostats in electric irons, refrigerators, and automatic kettles use bimetallic strips that bend upon heating to break the electrical circuit, regulating temperature.
Q.21	(B)	Flushing with large volumes of running water rapidly cools and washes away the acid. Never apply strong alkalis, as exothermic neutralization causes severe thermal burns.

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.22	(B)	Dilution of acid is highly exothermic. Adding acid slowly to a large volume of water allows the water to absorb the heat, preventing dangerous acid splattering.
Q.23	(B)	The wire gauze (often ceramic-centered) diffuses heat uniformly across the glass surface, preventing localized hot spots that shatter laboratory glassware.
Q.24	(B)	The non-luminous outer blue cone receives maximum oxygen, ensuring complete combustion and reaching temperatures up to 1500°C without leaving soot.
Q.25	(A)	Chemical splash goggles protect eyes from unexpected eruptions/splatters, while cotton lab coats protect skin and clothing from corrosive chemical damage.